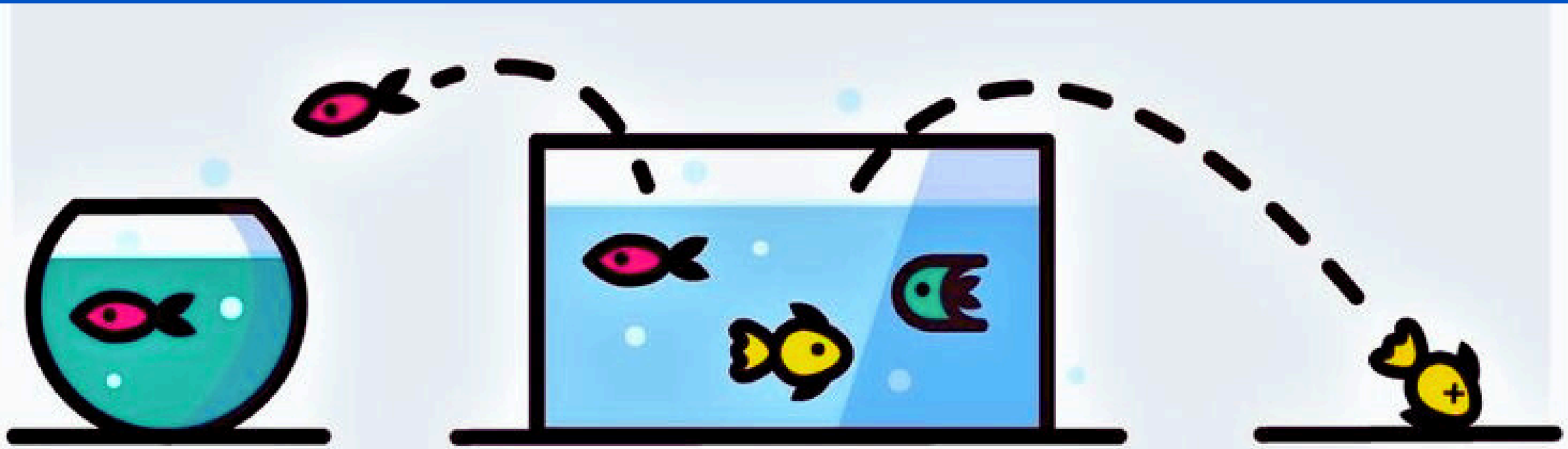


CUSTOMER CHURN USING DATA MINING

What is **Churn**?

Churn occurs when a customer leaves a company or cancels its service or move to another company.



PROBLEM DEFINITION

Customer **churn** represents one of the most critical challenges for **telecommunications companies,** as losing customers leads to significant:

- **Financial losses**
- **Higher Costs of acquiring new ones**.

Why Predict **Churn**?

- Detect churn risk **early**.
- Improve service quality
- Address **customer issues**
- **Increase** company **Revenues**.

Business Impact:

- Enhances **operational efficiency**
- **S**upports **better decision-making.**
- Strengthens **long-term customer loyalty.**
- **Attracting new customers** more effectively.
- Increases **competitive advantage.**



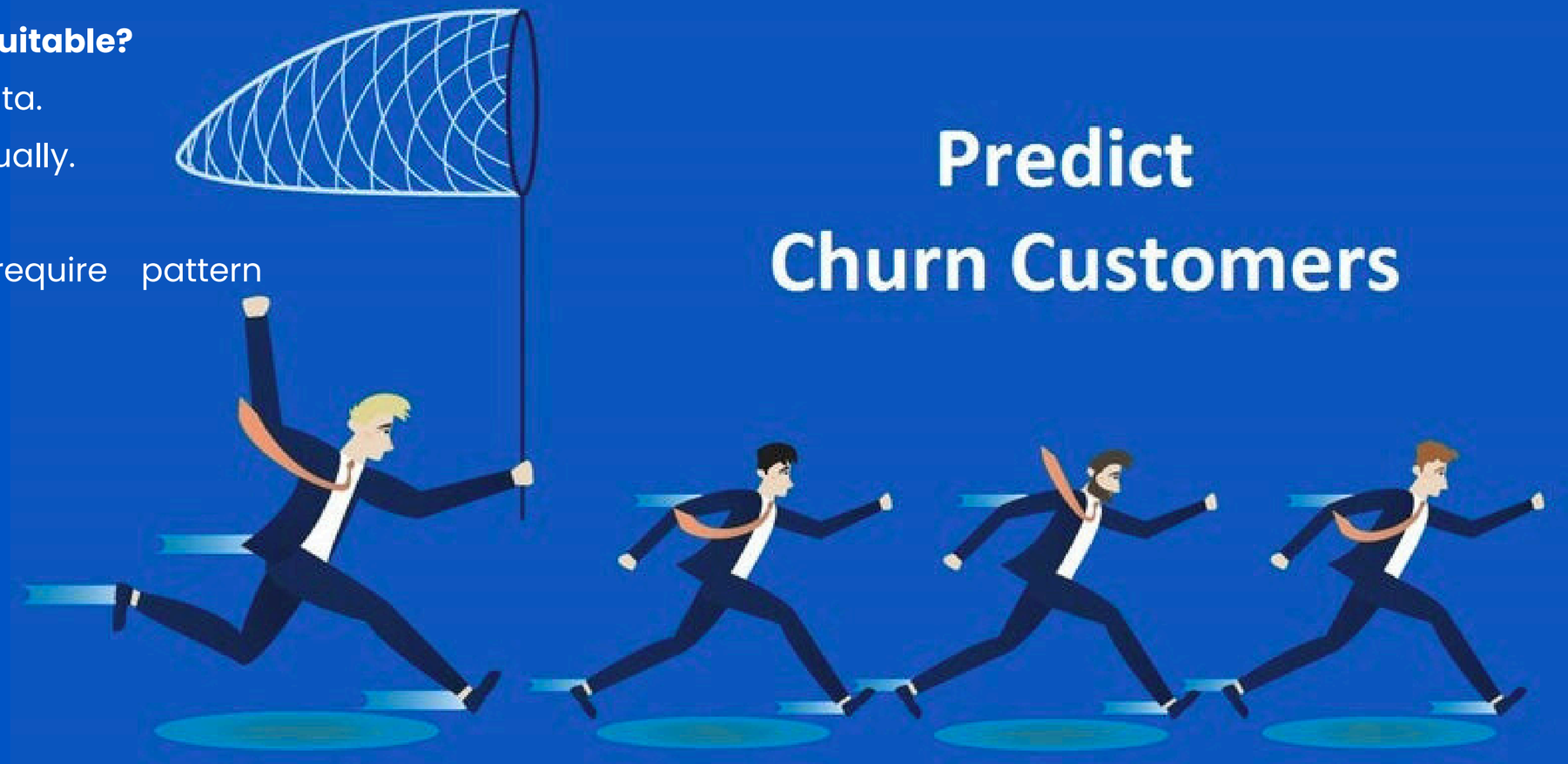
PROBLEM DEFINITION

Objectives:

- It is expected that a model will be developed to classify **customers who will leave the company**.

Why Data Mining Techniques Are Suitable?

- Large and complex customer data.
- Hidden patterns not visible manually.
- Predict churn accurately.
- Complex relationships that require pattern discovery.



ABOUT THE DATA

Data source:

Kaggle

About Data:

- **Telco churn data includes information about a fictitious telecom company that provided home phone and Internet services to 7,043 customers in California in the third quarter. It shows which customers left, stayed, or signed up for their service.**

	SeniorCitizen	tenure	MonthlyCharges	TotalCharges	Churn	gender_Male	Partner_Yes
0	0	1	29.85	29.85	0	False	True
1	0	34	56.95	1889.50	0	True	False
2	0	2	53.85	108.15	1	True	False
3	0	45	42.30	1840.75	0	True	False
4	0	2	70.70	151.65	1	False	False

ABOUT THE DATA

About the data set:

It contains:

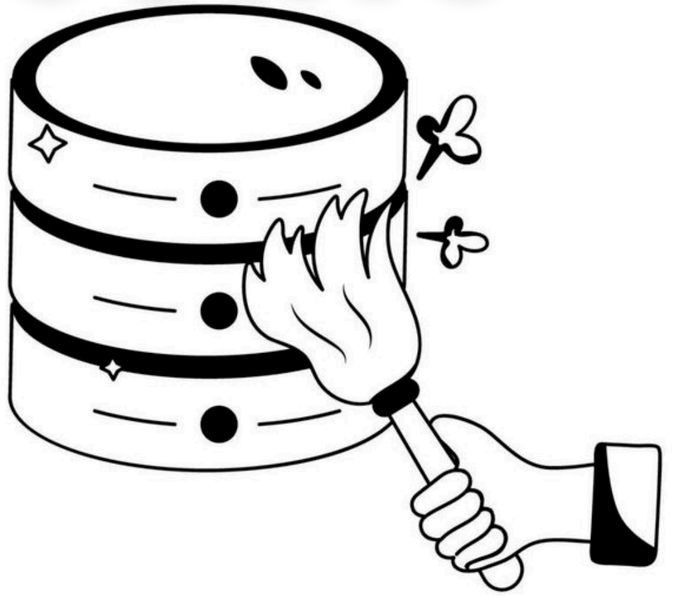
- **21 columns**

Types: (object ,float,Boolean,integer)

- **7043 rows.**



DATA PREPROCESSING



Remove unnecessary columns (Customer ID).

Transform features of type object into numerical features (float or integer).

Unknown or unseen classes are handled appropriately.



Use One-Hot Encoding for categorical features with two or more unordered classes.

For categorical object features with two classes(yes,no), apply Label Encoding or map values directly to binary values (0/1).



Use boxplots to detect outliers and remove or treat them if necessary.

Fill missing or null values using the mean, median, or mode, depending on the nature and distribution of the data.

Before

	customerID	gender	SeniorCitizen	Partner	Dependents	tenure	PhoneService	MultipleLines
0	7590-VHVEG	Female	0	Yes	No	1.0	No	No phone service
1	5575-GNVDE	Male	0	No	No	34.0	Yes	No
2	3668-QPYBK	Male	0	No	No	2.0	Yes	No
3	7795-CFOCW	Male	0	No	No	45.0	No	No phone service
4	9237-HQITU	Female	0	No	No	2.0	Yes	No

5 rows × 21 columns

After

	SeniorCitizen	Partner	Dependents	tenure	PhoneService	PaperlessBilling	MonthlyCharges	T
0	0	1	0	1.0	0	1	29.85	
1	0	0	0	34.0	1	0	56.95	
2	0	0	0	2.0	1	1	53.85	
3	0	0	0	45.0	0	0	42.30	
4	0	0	0	2.0	1	1	70.70	

5 rows × 42 columns

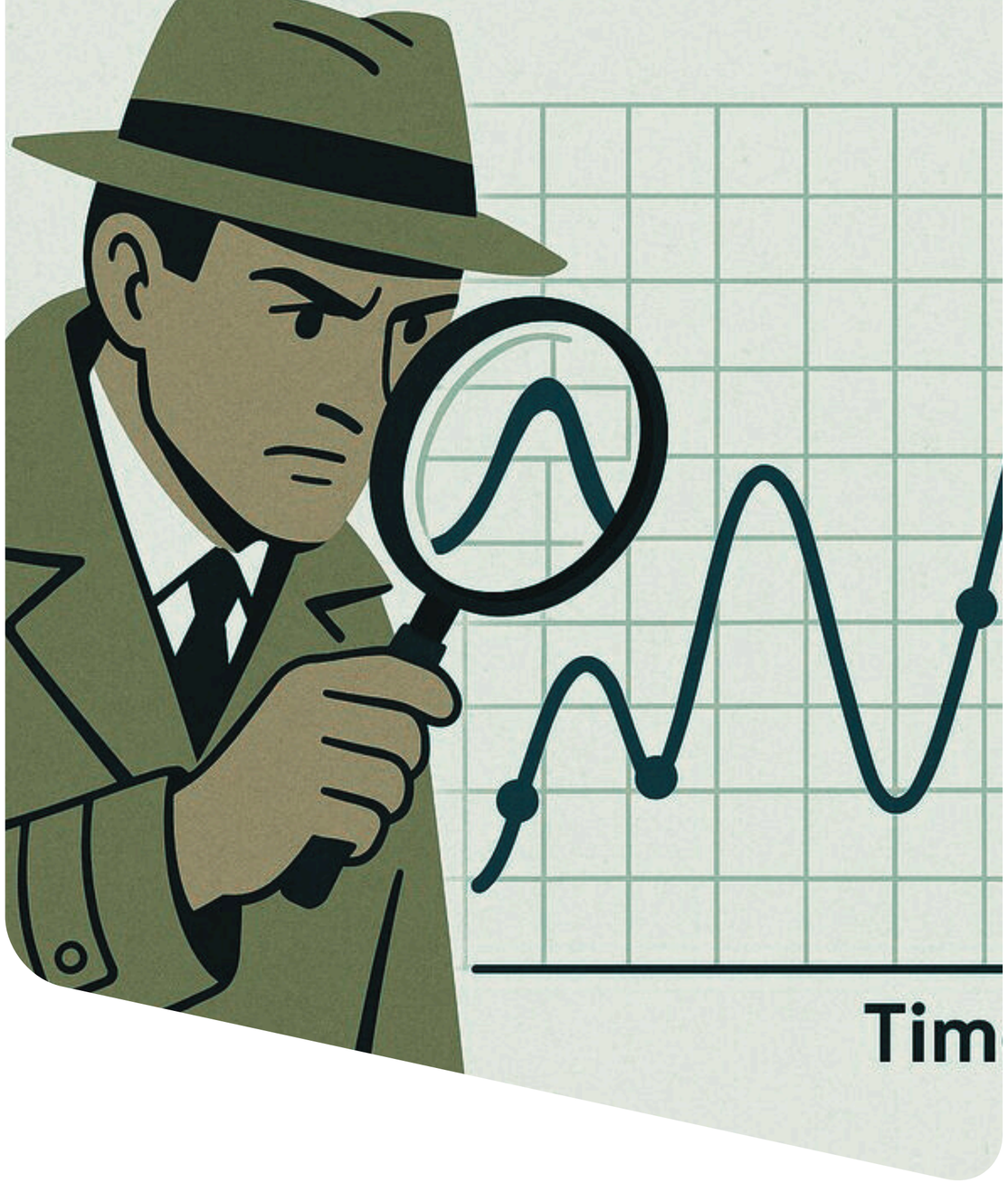
Why We Chose a Classification Model

We selected a classification model to predict whether customers will churn or stay. It's a binary problem with two clear outcomes, making decisions direct and actionable.

The benefits of using a classification model :

- Direct predictions: Clear "Yes" or "No" for each customer.
- Business impact: Focus resources on at-risk customers and avoid unnecessary incentives.
- Interpretability: Easy to explain results and integrate into business decisions.

Overall, the classification approach allowed us to turn data into clear actions, helping the company reduce churn and protect revenue



Why Logistic Regression

we used Logistic Regression because it is very suitable for our case: we need a binary prediction (whether a customer will leave the company or not), it provides the probability for each customer, and it is easy to interpret and use for decision-making.

In our model, 1 represents a customer predicted to churn (likely to leave), and 0 represents a customer predicted to stay. This is how the model turns probabilities into final decisions: if the predicted probability is above the threshold, it is classified as 1 (churn); if it is below, it is classified as 0 (non-churn)



EVALUATING MODEL PERFORMANCE

Understanding key metrics and their impact on business decisions

To properly evaluate a churn prediction model, we rely on multiple metrics, as each reveals a different facet of performance and influences business decisions in a distinct way.

First, we split the data into a training set and a test set. This ensures the model learns from one portion and is fairly evaluated on another, providing an accurate measure of its performance.

MEANINGFUL METRICS

Together, these metrics provide a complete picture—not only of how the model performs on data, but also of how its predictions will impact real business outcomes, from resource allocation to revenue preservation .

ACCURACY

PRECISION

RECALL

F1-SCORE

EVALUATING MODEL PERFORMANCE

Accuracy

A broad view of overall correctness but can be deceptive. In imbalanced churn datasets—where most customers stay—a model can score high accuracy while completely failing to identify those who leave.

Precision

A measures how accurate the model's predictions are when it predicts a customer will leave the company.
In our case (Churn): Shows, among the customers predicted to leave, how many actually left.
High precision means fewer false alarms, allowing us to focus retention efforts confidently and avoid wasting incentives on loyal customers.

Recall

measures the model's ability to identify customers who actually left the company. In churn prediction, it shows how many of the customers who truly churned were correctly predicted by the model.
A high recall means fewer churning customers are missed, which helps protect revenue, making recall one of the most important metrics in churn prediction

F1-Score

combines Precision and Recall into a single metric to evaluate overall model performance.
In churn prediction, it is used to choose a model that balances catching at-risk customers while avoiding too many false alerts. This balance helps achieve effective customer retention without wasting operational resources

THANK YOU



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